

Matrix Transforms

Domain, Codomain and Range

A function/transform/mapping has a **domain** and **codomain**: the domain specifies the possible inputs and the codomain specifies the possible outputs. The function maps each element of the domain to an **image** in the codomain.

The **range** is the set of all images: it is a subset of the codomain.

Matrix Transform

Defn. If the matrix A is $m \times n$, then the **matrix transform** $x \mapsto Ax$ has domain $\mathbb{R}^n$, codomain $\mathbb{R}^m$, and range some subset of $\mathbb{R}^m$.

For example, if A is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then the transform maps $(5, 3)$ to $(3, 5)$.

Some Matrix Transforms

⋈ projections, such as $P = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

⋈ shears, such as $S = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$

⋈ contractions/dilations, such as $C = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$

⋈ rotations, such as $R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

Summary

A transform maps each element of the domain to an image in the codomain. The range is the set of all images. A transform is onto if the range is the whole codomain, and one-to-one if every vector in the range is the image exactly once. Some matrix transforms have geometrical meaning.